Assignment 2

Use Python/R/Matlab coding – Do not directly use software for linear regression – You may check your answers (validate) by using a software after coding

Due Date: 6th April 2026 (No extensions permitted)

Question-1:

Reference: Kutner, M. H., C. J. Nachtschiem, J. Netner, Applied Linear Regression Models. 4th ed. New Delhi: McGraw Hill, 2004.

A researcher studied the effects of the charge rate and temperature on the life of a new type of power cell in a preliminary small-scale experiment. The charge rate (X_1) was controlled at three levels (0.6, 1.0, and 1.4 amperes) and the ambient temperature (X_2) was controlled at three levels (10, 20, 30°C). Factors pertaining to the discharge of the power cell were held at fixed levels. The life of the power cell (Y) was measured in terms of the number of discharge-charge cycles that a power cell underwent before it failed. The data obtained in the study are given below:

Lifecycle (Y)	**Charge Rate (X_1)**	**Temperature (X_2)**
144	0.6	10
89	1.0	10
59	1.4	10
278	0.6	20
167	1.0	20
141	1.0	20
164	1.0	20
129	1.4	20
259	0.6	30
245	1.0	30
214	1.4	30

The researcher was not sure about the nature of the response function in the range of the factors studied. Hence, the researcher decided to fit the second-order polynomial regression model as follows:

$$Y_i = \beta_0 + \beta_1 x_{\{i1\}} + \beta_2 x_{\{i2\}} + \beta_{\{11\}} x_{\{i1\}}^2 + \beta_{\{22\}} x_{\{i2\}}^2 + \beta_{\{12\}} x_{\{i1\}} x_{\{i2\}} + \varepsilon_i \qquad (1)$$

for which the response function is:

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{\{11\}} x_1^2 + \beta_{\{22\}} x_2^2 + \beta_{\{12\}} x_1 x_2 \qquad (2)$$

Write a code (using R/Python/MATLAB) to find the following:

i. Create a matrix "**X**" for raw data

ii. Create matrix "**x**" for data such that $x = \frac{X-\mu}{Step\ size}$

iii. Create a matric "z" such that $z = \frac{X-\mu}{\max(X)-\min(X)}$

iii. Find regression coefficients for the three different cases (X, x, and z).

iv. Find residual sum of squares, regression sum of squares, total sum of squares. Use both Matrix and summation approaches and compare them. Do the values change depending upon the transformation?

v. Find the variance-covariance matrix in all cases and compare the matrices for the transformed(x,z) and non-transformed (X) cases. Which is better in terms of precision of the regression coefficient estimates?

vi. Find R^2, adj. R^2, PRESS, R^2_{PRESS} for all the cases.

Write code for finding PRESS and R^2_{PRESS} in all the three cases.

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Question 2

A Three-Component Mixture Cornell (2002) describes a mixture experiment in which three components—polyethylene (x_1), polystyrene (x_2), and polypropylene (x_3)—were blended to form fiber that will be spun into yarn for draperies. The product developers were interested in only the pure and binary blends of these three materials. The response variable of interest is yarn elongation in kilograms of force applied. A {3, 2} simplex lattice design is used to study the product. The design and the observed responses are shown Notice that replicate observations are run, with two replicates at each of the pure blends and three replicates at each of the binary blends. The error standard deviation can be estimated from these replicate observations as $\hat{\sigma} = 0.85$.

$$\hat{y}(\mathbf{x}) = b_1x_1 + b_2x_2 + b_3x_3 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3$$

Observed Yarn Elongation Values

Design Point	Component Proportions x_1	x_2	x_3	Observed Elongation Values (y_u)	Average Elongation Value ($\bar{y}$)
1	1	0	0	11.0, 12.4	11.7
2	$\frac{1}{2}$	$\frac{1}{2}$	0	15.0, 14.8, 16.1	15.3
3	0	1	0	8.8, 10.0	9.4
4	0	$\frac{1}{2}$	$\frac{1}{2}$	10.0, 9.7, 11.8	10.5
5	0	0	1	16.8, 16.0	16.4
6	$\frac{1}{2}$	0	$\frac{1}{2}$	17.7, 16.4, 16.6	16.9

Use your own Matlab/Python code – you may modify the previously given code for this problem

a) How is this mixture regression problem different from conventional linear regression?
b) Give the $\boldsymbol{X}$ matrix, $\hat{\beta}$ and **Y** matrices/vectors
c) Estimate the parameters
d) Find the total, regression and error sum of squares
e) What is the source of the error? Why is it called pure error?
f) Estimate the total, regression and error sum of squares using direct matrix manipulations and conventional summation methods
g) Find the variance covariance matrix
h) Find the R^2, adj. R^2, PRESS and R2 Press

i) Find the settings that will maximize Yarn Elongation (write constrained optimization code for this)
j) Find the settings that will minimize Yarn Elongation (write constrained optimization code for this)
k) Confirm your results using a standard software like Polymath, Minitab, or JMP

[2+3+6+2+6+2+8+4+4+3]

Question 3

The programming example discussed in class is to be solved using the code. Find the parameters of the 1-predictor logistic regression model and compare the predicted probabilities with the actual binary inputs. Use Matlab/Python coding.

Compare your answers with JMP/Minitab if possible.

[15]

Question 4

The dark chocolate example data is given below. Now develop the logistic models for the given data using product of Bernoulli outcomes

a) Only income as the predictor
b) Both income as well as gender as predictors
c) Compare your answers with JMP/Minitab if possible.

Data is given in terms of sl. No., income (in thousands of euros), gender (0-female, 1-male), decision (0-not urchased, 1- purchased)

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Table 2. Data for Programming Experience versus Task Completion

Experience	y	y_Pred
14	0	0.310262
29	0	0.835263
6	0	0.109996
25	1	0.726602
18	1	0.461837
4	0	0.08213
18	0	0.461837
12	0	0.245666
22	1	0.620812
6	0	0.109996
30	1	0.856299
11	0	0.21698
30	1	0.856299
5	0	0.095154
20	1	0.542404
13	0	0.276802
9	0	0.1671
32	1	0.891664
24	0	0.693379
13	1	0.276802
19	0	0.502134
4	0	0.08213
28	1	0.811825
22	1	0.620812
8	1	0.145815

Table 2. Data for Dark Chocolate Buying Trend in terms of Income and gender

Data_Expt = [

1 2.53 0 1

2 2.37 1 0

3 2.72 1 1

4 2.54 0 0

5 3.20 1 1

6 2.94 0 1

7 3.20 0 1

8 2.72 1 1

9 2.93 0 1

10 2.37 0 0

11 2.24 1 1

12 1.91 1 1

13 2.12 0 1

14 1.83 1 1

15 1.92 1 1

16 2.01 0 0

17 2.01 0 0

18 2.23 1 0

19 1.82 0 0

20 2.11 0 0

21 1.75 1 1

22 1.46 1 0

23 1.61 0 1

24 1.57 1 0

25 1.37 0 0

26 1.41 1 0

27 1.51 0 0

28 1.75 1 1

29 1.68 1 1

30 1.62 0 0];

Data also given as excel file in book resources and image reproduced below

Person i	Income [1000 EUR]	Gender 0 = f, 1 = m	Buy 1 = yes, 0 = no
1	2.53	0	1
2	2.37	1	0
3	2.72	1	1
4	2.54	0	0
5	3.20	1	1
6	2.94	0	1
7	3.20	0	1
8	2.72	1	1
9	2.93	0	1
10	2.37	0	0
11	2.24	1	1
12	1.91	1	1
13	2.12	0	1
14	1.83	1	1
15	1.92	1	1
16	2.01	0	0
17	2.01	0	0
18	2.23	1	0
19	1.82	0	0
20	2.11	0	0
21	1.75	1	1
22	1.46	1	0
23	1.61	0	1
24	1.57	1	0
25	1.37	0	0
26	1.41	1	0
27	1.51	0	0
28	1.75	1	1
29	1.68	1	1
30	1.62	0	0

Question 5

In a study of the effectiveness of coupons offering a price reduction on a given product, 1,000 homes were selected at random. A packet containing advertising material and a coupon for the product were mailed to each home. The coupons offered different price reductions (5, 10, 15, 20, and 30 dollars), and 200 homes were assigned at random to each of the price reduction categories. The predictor variable X in this study is the amount of price reduction, and the response variable Y is a binary variable indicating whether or not the coupon was redeemed within a six-month period.

Table 14.2 contains the data for this study. X_j denotes the price reduction offered by

a coupon, n_j the number of households that received a coupon with price reduction X_j , $Y_{.j}$ the number of these households that redeemed the coupon, and p_j the proportion of households receiving a coupon with price reduction X_j that redeemed the coupon. The logistic regression model is to be fitted by a logistic regression model

$$\pi_j = \frac{\exp(\beta_0 + \beta_1 X_j)}{1 + \exp(\beta_0 + \beta_1 X_j)}$$

Level j	(1) Price Reduction X_j	(2) Number of Households n_j	(3) Number of Coupons Redeemed $Y_{.j}$	(4) Proportion of Coupons Redeemed p_j
1	5	200	30	.150
2	10	200	55	.275
3	15	200	70	.350
4	20	200	100	.500
5	30	200	137	.685

Compare the predicted probabilities with actual proportions observed.

[15]

Question 6

You have done Binomial distribution of grouped data involving repeated predictors. You estimated parameters using this data set. Now ungroup the dataset but use

repeated averaged income predictor values instead of actual income values. The response values remain as given. Hence, the dataset is given as follows

Income	**y**
2.717	1
2.717	0
2.717	1
2.717	0
2.717	1
2.717	1
2.562	1
2.562	1
2.562	1
2.562	0
2.562	1
2.562	1
2.02	1
2.02	1
2.02	1
2.02	0
2.02	0
2.02	0
1.72	0
1.72	0
1.72	1
1.72	0
1.72	1
1.72	0
1.556667	0
1.556667	0
1.556667	0
1.556667	1
1.556667	1
1.556667	0

Now use Bernoulli trials based MLE method the find the logistic regression model parameters. Compare them with the parameters estimated from Binomial trials based MLE model parameters using grouped data of repeated predictors.

[10]

Question 7

Spread of Dengue due to mosquitoes in different sectors (1 and 2) of a town was analyzed. The sectors may be coded as 0 and 1. The economic status was also taken into account and this had three levels (lower/middle/upper). These were coded in terms of indicator variables as

Upper:	0	0
Middle:	1	0
Lower:	0	1

The response variable Y was coded 1 for disease occurrence and 0 if not.

The data for 98 individuals in the sample are given below. Estimate the model parameters using MLE criterion.

For downloading data use the following link

http://www.cnachtsheim-text.csom.umn.edu/Kutner/Chapter%2014%20Data%20Sets/CH14TA03.txt

City sector is also a categorical variable. Since there were only two sectors in the study, one indicator variable *(X_4)* was used, defined so that $X_4 = 0$ for sector 1 and $X_4 = 1$ for sector 2. The reason why the upper socioeconomic class was chosen as the reference class (i.e., the class for which the indicator variables X_2 and X_3 are coded 0) is that itwas expected that this classwould have the lowest disease rate among the socioeconomic classes. By making this class the reference class, the odds ratios associated with regression coefficients β_2 and $\beta 3$ would then be expected to be greater than 1, facilitating their interpretation. For the same reason, sector 1, where the epidemic was less severe, was chosen as the reference class for the sector indicator variable X_4. Note the use of the indicator variables as just explained for the two categorical variables. The primary purpose of the study was to assess

the strength of the association between each of the predictor variables and the probability of a person having contracted the disease.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$